

# Yonghoon Jeong

Full-time Lecturer, Department of Cyber Science  
Researcher, Institute of Naval Research  
Republic of Korea Naval Academy

## Contact Information

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## Research Interests

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Wireless Communication Systems  
Waveform Design and Signal Processing  
Low-Power Wide-Area Networks (LPWANs)  
Underwater Backscatter Systems  
Ocean IoT

## Education

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| <b>M.S. in Electrical and Computer Engineering</b>                                                                                 | 2024 – 2026 |
| <ul style="list-style-type: none"><li>Seoul National University, Seoul, South Korea</li><li>Advisor: Prof. Saewoong Bahk</li></ul> |             |
| <b>B.S. in Electronics and Communications Engineering</b>                                                                          | 2018 – 2023 |
| <ul style="list-style-type: none"><li>Kwangwoon University, Seoul, South Korea</li></ul>                                           |             |

## Professional Experience

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| <b>Republic of Korea Naval Academy (ROKNA)</b> , Changwon, South Korea                                                                                                                    | 2026 – Present |
| <ul style="list-style-type: none"><li>Full-time Lecturer, Department of Cyber Science</li><li>Researcher, Institute of Naval Research</li></ul>                                           |                |
| <b>Seoul National University (SNU)</b> , Seoul, South Korea                                                                                                                               | 2024 – 2026    |
| <ul style="list-style-type: none"><li>Researcher, Network Lab (NETLAB), Institute of New Media and Communications (INMC)</li></ul>                                                        |                |
| <b>Electronics and Telecommunications Research Institute (ETRI)</b> , Seongnam, South Korea                                                                                               | 2023           |
| <ul style="list-style-type: none"><li>Researcher, AI-SoC Infrastructure Research Section</li></ul>                                                                                        |                |
| <b>Korea Advanced Institute of Science and Technology (KAIST)</b> , Daejeon, South Korea                                                                                                  | 2022           |
| <ul style="list-style-type: none"><li>Research Intern, Unmanned Systems Research Group (USRG), School of Electrical Engineering (EE)</li><li>Advisor: Prof. David Hyunchul Shim</li></ul> |                |

## Publications

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### International Journals

- **Yonghoon Jeong**, Youngwook Son, and Saewoong Bahk, “On the Interference Resilience of Non-Linear Chirp Modulation for LoRa: An Analytical Insight into Energy Scattering Effect”, *IEEE Communications Letters*, vol. 30, pp. 2238–2242, Jun. 2026.
- Dongrakk Choi, Yubin Choi, **Yonghoon Jeong**, Jeongyeup Paek, and Saewoong Bahk, “D<sup>2</sup>-PLC: Holistic Design and Implementation of High-Datarate Duplex DC Power Line Communication Network”, *IEEE Internet of Things Journal*, vol. 12, pp. 45325–45340, Nov. 2025.
- **Yonghoon Jeong** and Youngseok Choi, “Diffusion-Phys: Noise-Robust Heart Rate Estimation from Facial Videos via Diffusion Models”, *Biomedical Engineering Letters*, vol. 15, pp. 575–585, Apr. 2025.
- “Towards Optimized Chirp Modulation for LoRa under Severe Doppler Effects”, in preparation.
- “Underwater Backscatter Communications with Doppler-Resilient Chirp Waveforms”, in preparation.

## International Conferences

- Dongrak Choi, **Yonghoon Jeong**, Yubin Choi, and Saewoong Bahk, “A Programmable High-Throughput Duplex DC-PLC Testbed for Power and Data Integration”, *IEEE ICNP 2025*, Seoul, Korea, Oct. 19–22, 2025.

## Domestic Conferences

- **Yonghoon Jeong**, Saewoong Bahk, “A Study on Direct Current Power Line Communication (DC-PLC) with Applications in Automotive, Maritime, and Aviation,” *The Fall Conference of the Korea Institute of Communications and Information Sciences (KICS)*, Gyeongju, Korea, November 20-22, 2024.
- **Yonghoon Jeong**, “Development of Non-contact Heart Rate Measurement Technology Based on Facial Video,” *The Conference of the Korean Institute of Information Scientists and Engineers (KIISE)*, Jeju, Korea, June 23-25, 2021.

## Awards / Scholarships

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|------------------------------------------------------------------------------------------------------------------|-----------|
| <b>Scholarship Recipient</b>                                                                                     | 2024–2025 |
| <ul style="list-style-type: none"> <li>• Brain Korea 21 (BK21), National Research Foundation of Korea</li> </ul> |           |
| <b>Scholarship Recipient</b>                                                                                     | 2022–2023 |
| <ul style="list-style-type: none"> <li>• Korea Scholarship Foundation for the Future Leaders (KOSFFL)</li> </ul> |           |
| <b>Excellent Paper Awards</b>                                                                                    | 2021      |
| <ul style="list-style-type: none"> <li>• Korea Computer Conference 2021</li> </ul>                               |           |
| <b>University President’s Award</b>                                                                              | 2020      |
| <ul style="list-style-type: none"> <li>• Seoul Capstone Design, Seoul</li> </ul>                                 |           |
| <b>Scholarship Recipient</b>                                                                                     | 2020      |
| <ul style="list-style-type: none"> <li>• Samsung Scholarship Foundation</li> </ul>                               |           |
| <b>Dean’s List</b>                                                                                               | 2020      |
| <ul style="list-style-type: none"> <li>• Highest GPA</li> </ul>                                                  |           |

## Programming Skills

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**Programming Languages:** C, C++, Python, MATLAB

**Frameworks & Libraries:** ROS, ROS2, TensorFlow, PyTorch

**Simulation Tools:** MATLAB/Simulink, ns-3

**Development Tools:** Linux, Git, Docker

## Teaching Experience

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|---------------------------------------------------------------------------|------|
| <b>Creative Engineering Design</b> , Prof. Sae-woong Bahk                 | 2024 |
| <b>Electronic Circuits Experiment 1</b> , Emeritus Prof. Young-chul Chung | 2023 |
| <b>Electronic Circuits Experiment 2</b> , Emeritus Prof. Young-chul Chung | 2023 |
| <b>Electronic Circuits Experiment 1</b> , Prof. Young-seok Choi           | 2022 |